FA2 REFLECTION

Biology Laboratory Record

1. Title & Objective

- **Experiment Title:** Longitudinal Section (L.S.) of a Mammalian Kidney
- **Objective:** To study and document the external and internal morphological features of a mammalian kidney (sheep or goat).

2. Scientific Principle & Background

- Excretion is the biological process of removing toxic metabolic nitrogenous wastes and excess fluid from the body to maintain internal osmotic balance (homeostasis).
- The kidneys are the primary vertebrate excretory organs, responsible for filtering blood, balancing electrolytes, and producing urine.

3. Materials Required

- Fresh specimen of a mammalian kidney (sheep or goat)
- Sharp surgical blade or scalpel
- Dissection tray
- Jug/bottle of water for washing
- Hand lens or magnifying glass

4. Step-by-Step Procedure

- **Cleaning:** Collect a fresh kidney specimen and wash it thoroughly under running water to clear away any residual blood clots and clean the outer surface.
- **External Examination:** Place the kidney on a dissection tray. Observe its shape, color, surface texture, and outer coverings, and note down the findings.
- **Sectioning:** Using a sharp, clean surgical blade, carefully cut the kidney vertically along its longitudinal axis (L.S.) into two equal halves.

- **Internal Examination:** Rinse the cut halves gently with water to clear any loose tissue, and use a hand lens to examine the internal zones, regions, and tubes.

5. Observations

- **External Features:**
  - The kidneys are characteristically bean-shaped structures with a reddish-brown color.
  - Each kidney has a convex outer border and a concave inner border.
  - The concave inner side features a depression known as the hilum (or fissure), through which the renal artery enters, the renal vein exits, and the ureter leaves the kidney.
- **Internal Features (L.S.):**
  - When viewed internally, the L.S. of the kidney clearly displays two distinct regional zones.
  - The outer zone is darker in color and is called the cortex.
  - The inner zone is paler in color and is called the medulla, which consists of conical tissue masses called renal pyramids.

6. Detailed Anatomical Findings & Functional Significance

- Each kidney is composed of more than one million microscopic, highly convoluted tubular functional units called nephrons (or uriniferous tubules).
- Nephrons are the true structural and functional units of the kidney; they perform ultrafiltration, selective reabsorption, and tubular secretion to convert blood plasma into urine.
- The urine collected from the nephrons drains into the renal pelvis and moves down through the ureter to the urinary bladder for temporary storage before elimination.

7. Precautions & Learning Outcomes

- Always use a sharp blade to obtain a clean, smooth cut without crushing delicate internal tissues.
- Maintain proper hygiene by washing hands and disinfecting tools thoroughly after handling raw animal tissue.
- This practical provides a clear understanding of mammalian renal macro-anatomy and how structural zonation supports physiological blood filtration.